

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 11



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Recap of Previous Lecture



Topic

Least Upper Bound



Topic

Greatest Lower Bound



Topic

E.g. of Least Upper Bound & Greatest Lower Bound



Topics to be Covered



Topic

Minimal / Maximal elements in a POSET



Topic

Minimum(Least) element of POSET



Topic

Maximum(Greatest) element of POSET



Topic

Lattice





Topic : Minimal elements in a POSET

There may be more than one minimal element

R ← it can be any partial order relation

Let $(A, \leq)$ be a POSET

An element $x \in A$ is called a minimal element of $\text{POSET}(A, \leq)$ if there exists no element $y \in A$ other than x itself, such that $y \leq x$.

i.e. $y R x$

i.e., if there is no element 'y' other than 'x' itself which is at the lower side of x, then 'x' is a minimal element of the POSET.

eg:- $A = \{2, 3, 4, 6, 12\}$

and $(A, \div)$ is a POSET.

find all minimal elements of the POSET.

* There is no element that divides '2' other than '2' itself

∴ '2' is a minimal element

* There is no element that divides 3 other than 3 itself

∴ '3' is a minimal element

$$(2, 4) \in \div$$

$$(2, 6) \in \div$$

$$(3, 12) \in \div$$

$\underbrace{(2, 4), (2, 6), (3, 12)}_{4, 6 \text{ \& } 12 \text{ are not minimal}}$



Topic : Maximal elements in a POSET

Let $(A, \leq)$ be a POSET

An element $x \in A$ is called a maximal element of $\text{POSET}(A, \leq)$ if there exists no element $y \in A$ other than x itself, such that $x \leq y$.

$$x R y$$

If there is no element 'y', other than 'x' itself which is at the upper side of x, then x is the maximal element of POSET.

eg:- $A = \{2, 3, 4, 6, 15\}$

and $(A, \div)$ is a POSET.

Find all maximal elements of the POSET.

* 4, 6, and 15 are the maximal elements of the POSET.

eg:- $A = \{2, 3, 4, 6, 12\}$

and $(A, \div)$ is a POSET.

Find all maximal elements of the POSET.

→ '12' is the only maximal element of POSET.

→ Whereas, 2 & 3 are minimal elements.

eg:- $A = \{1, 2, 3, 4, 6, 12\}$
and $(A, \div)$ is a POSET.

find all minimal and maximal elements of POSET.

* '1' is the only minimal element

* '12' is the only maximal element



Topic : Minimum element of a POSET

Let $(A, \leq^R)$ be a POSET

An element $m \in A$ is called a minimum element of $\text{POSET}(A, \leq^R)$

if and only if

$$m \leq a, \forall a \in A$$

$$(m^R a, \forall a \in A)$$

'm' at lower side of all elements of set A.

eg:- $A = \{2, 3, 4, 6, 15\}$

and $(A, \div)$ is a POSET.

Find minimum element of POSET.

→ There is no element in the set that divides all elements of the set.

i.e., No element which is at the lower side of all elements of the set

Hence, minimum element does not exist in the above POSET.

Note:-

If there exists two or more minimal elements in a POSET, then minimum element can never exist in that POSET.

eg:- $A = \{1, 2, 3, 4, 6, 12\}$

and $(A, \div)$ is a POSET.

Find minimum element of POSET.

'1' divides all the elements of the set,
∴ '1' is at the lower side of all the elements
of the set

Hence, '1' is the minimum element of the POSET.

Note: '1' is also the minimal element

Note:-

If there exists a unique minimal element in the POSET, then that element will also be the minimum element of POSET.

Note:

Minimum element if exists, then it is always unique.



Topic : Maximum element of a POSET

Let $(A, \leq^R)$ be a POSET

An element $n \in A$ is called a maximum element of $\text{POSET}(A, \leq^R)$

if and only if

$$a \leq n, \forall a \in A$$

$$a^R n, \forall a \in A$$

i.e., 'n' is at the upper side of all element of the set.

eg:- $A = \{2, 3, 4, 6, 15\}$

and $(A, \div)$ is a POSET.

Find maximum element of POSET.

→ There is no element in the set that can be divided by all elements of the set.

i.e., No element which is at the upper side
of all elements of the set

Hence, maximum element does not exist in the
above POSET.

Note:-

If there exists two or more maximal elements in a POSET, then maximum element can never exist in that POSET.

eg:- $A = \{1, 2, 3, 4, 6, 12\}$

and $(A, \div)$ is a POSET.

Find maximum element of POSET.

* '12' can be divided by all the elements of the set.

∴ '12' is at the upper side of all elements of set

Hence, '12' is the maximum element of the POSET.

Note: '12' is also the maximal element

Note:-

If there exists a unique maximal element in the POSET, then that element will also be the maximum element of POSET.

Note:

Maximum element if exists, then it is always unique.

Note:-

In a POSET, minimum and/or maximum element need not exist, but if exists then they will be unique.



Topic : NOTE



* Let $\mathbb{Z}$ is the set of all integers, then

$(\mathbb{Z}, \leq)$ is a POSET.
less than or equal

* No minimal element exists in the POSET
And hence no minimum element.

* No maximal element exists in the POSET
And hence no maximum element.



Topic : NOTE



→ Let N is the set of all natural numbers, then

$(N, \leq)$ is a POSET.

↖ less than or equal

→ '1' is the only minimal element of the set
and it is also the minimum element

→ No maximal element exists in the POSET
and hence no maximum element

→ If there is no minimal element in the POSET and/or no maximal element in the POSET, then Set is guaranteed to be an infinite set.

→ But, if Set is an infinite set then minimal and maximal elements may or may not exist.



Topic : NOTE



eg. Let $A = \{x \mid x \in \mathbb{R} \text{ and } 0 \leq x \leq 1\}$

it is an infinite set

→ Here $(A, \leq)$ is a POSET where

'0' is the minimal element

& '1' is the maximal element

Note :- ① If set is a finite set, then in
a POSET w.r.t. that set.

there will be at least one minimal element
and there will be at least one maximal element

② In a POSET {may or may not be finite}
there can be at most one minimum element
and there can be at most one maximum element

Topic : Lattice

Note:-

In a POSET, lub and/or glb need not exist for every pair of elements of the set



Topic : Join Semi Lattice

* A POSET in which least upper bound exists for every pair of elements of the set is called Join - semi lattice

{ glb, may or may not }
exist for every pair
of elements.



Topic : Meet Semi Lattice

* A POSET in which greatest lower bound exists for every pair of elements of the set is called a Meet semi lattice.

{ lub may or may not exist
for every pair of elements }



Topic : Lattice



→ A POSET in which both least upper bound (lub) as well as greatest lower bound (glb) exists for every pair of elements of the set is called a lattice

* A POSET that is a Join semi lattice as well as a Meet semi lattice is called a lattice

eg ① $A_1 = \{2, 3, 4, 6, 15\}$ (i) Check if lub exists for Every pair of elements
and $(A_1, \div)$ is a POSET (ii) Check if glb exists for Every pair of elements

② $A_2 = \{2, 3, 4, 6, 12\}$ (i) Check if lub exists for Every pair of elements
and $(A_2, \div)$ is a POSET (ii) Check if glb exists for Every pair of elements

③ $A_3 = \{1, 2, 3, 4, 6, 15\}$ (i) Check if lub exists for Every pair of elements
and $(A_3, \div)$ is a POSET (ii) Check if glb exists for Every pair of elements

④ $A_4 = \{1, 2, 3, 4, 6, 12\}$ (i) Check if lub exists for Every pair of elements
and $(A_4, \div)$ is a POSET (ii) Check if glb exists for Every pair of elements

eg ① $A_1 = \{2, 3, 4, 6, 15\}$ (i) Check if lub exists for Every pair of elements and $(A_1, \div)$ is a POSET (ii) Check if glb exists for Every pair of elements

(i) $\text{lub}(4, 6) = \text{does not exist}$

↳ There exists some pair of elements for which lub does not exist

(ii) $\text{glb}(2, 3) = \text{does not exist}$ ↳ Hence POSET $(A_1, \div)$ is not a Join semi lattice

↳ There exists some pair of elements for which glb does not exist

↳ Hence POSET $(A_1, \div)$ is not a Meet semi lattice

Q2) $A_2 = \{2, 3, 4, 6, 12\}$ (i) Check if lub exists for Every pair of elements
and $(A_2, \div)$ is a POSET (ii) Check if glb exists for Every pair of elements

(i) $\rightarrow$ We can observe that lub exist for every pair of element of the set, Hence POSET $(A_2, \div)$ is a Join semi lattice

(ii) $\text{glb}(2, 3) = \text{does not exist}$

$\hookrightarrow$ Hence, POSET $(A_2, \div)$ is not a Meet semi lattice

eg ③ $A_3 = \{1, 2, 3, 4, 6, 15\}$ (i) Check if lub exists for Every pair of elements
and $(A_3, \div)$ is a POSET (ii) Check if glb exists for Every pair of elements

(i) $\text{lub}(4, 15) = \text{does not exist}$

↳ Hence $\text{POSET}(A_3, \div)$ is not a Join Semi lattice

(ii) But we can observe that glb exists for

Every pair of elements

↳ Hence $\text{POSET}(A_3, \div)$ is a Meet Semi lattice

eg ④ $A_4 = \{1, 2, 3, 4, 6, 12\}$ (i) Check if lub exists for Every pair of elements
And $(A_4, \div)$ is a POSET (ii) Check if glb exists for Every pair of elements

* (i) Lub exist for Every pair of elements

↳ Hence POSET $(A_4, \div)$ is
a Join semi lattice

- (ii) glb exist for Every pair of elements

↳ Hence POSET $(A_4, \div)$ is
a Meet semi lattice

} POSET $(A_4, \div)$ is
a Join semi lattice,
as well as a
Meet semi lattice,
∴ $(A_4, \div)$ is
a lattice

Note:- If 'n' is any positive integer, then

" D_n " is defined as a set of all
Positive divisors of 'n'.

ie, $D_1 = \{1\}$ $D_4 = \{1, 2, 4\}$ $D_{12} = \{1, 2, 3, 4, 6, 12\}$
 $D_2 = \{1, 2\}$ $D_5 = \{1, 5\}$ $\vdots$
 $D_3 = \{1, 3\}$ $D_6 = \{1, 2, 3, 6\}$ $\vdots$

★ Note:-

① If D_n is the set of all positive divisors of a positive integer 'n', then

$(D_n, \div)$ is always a lattice.

② If 'A' is any finite set, and $P(A)$ is the Power set of set A, then

$(P(A), \subseteq)$ is always a lattice

③ If A is any set of real numbers, then

$(A, \leq)$ is always a lattice



Topic : Lattice



A lattice is denoted by $[L, \vee, \wedge]$

Where, L = Underlying set w.r.t. POSET.

$\vee$ = least upper bound w.r.t.
Partial order relation of POSET

$\wedge$ = greatest lower bound w.r.t.
Partial order relation of POSET.

Properties w.r.t. Lattice

Let $[L, \vee, \wedge]$ be a lattice, then following properties will always hold true in a lattice

① Idempotent Law: (i) $a \vee a = a, \forall a \in L$

(ii) $a \wedge a = a, \forall a \in L$

② Commutative Law (i) $a \vee b = b \vee a, \forall a, b \in L$

(ii) $a \wedge b = b \wedge a, \forall a, b \in L$

③ Associativity Law (i) $(a \vee b) \vee c = a \vee (b \vee c), \forall a, b, c \in L$

(ii) $(a \wedge b) \wedge c = a \wedge (b \wedge c), \forall a, b, c \in L$

④ Absorption Law (i) $a \vee (a \wedge b) = a, \forall a, b \in L$

(ii) $a \wedge (a \vee b) = a, \forall a, b \in L$

These four properties will always hold true in any lattice

Note:- Distributive Property need not hold true for all lattices

↓

i.e. (i) $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$, $\forall a, b, c \in L$
and (ii) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$, $\forall a, b, c \in L$

← If above distributive property holds true in a lattice, then that lattice is called a distributive lattice

→ If distributive property does not hold true for at least one combination of a, b & c , then lattice is not a distributive lattice



2 mins Summary



Topic

Minimal / Maximal elements in a POSET ✓

Topic

Minimum(Least) element of POSET ✓

Topic

Maximum(Greatest) element of POSET ✓

Topic

Lattice ✓

THANK - YOU